Smit Limbasiya

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Profile

B.Tech student in Information and Communication Technology at Dhirubhai Ambani University with a strong interest in Software Engineering, Generative AI, and Machine Learning. Skilled in developing AI-powered applications, integrating LLMs, and building backend systems. Focused on applying technical knowledge to solve real-world problems.

Education

Dhirubhai Ambani University — Gandhinagar, Gujarat
B.Tech in Information and Communication Technology — *2023 – Present*

Aashadeep IIT (GHSEB) — Surat, Gujarat
Higher Secondary Education (Class XII) — *2022 – 2023*

Projects

CapitalUp – Real-Time Broker Trading Platform | *Node.js, Express.js, PostgreSQL, Redis, Kafka* GitHub

- **Trading & Matching Engine:** Engineered an in-memory price-time priority matching engine using Red-Black Trees for price levels, Doubly Linked Lists for FIFO order queues, and HashMap-based order lookup; supporting Market, Limit, IOC, FOK orders with partial fills, cancellation and deterministic per-symbol order processing.
- **Performance Engineering:** Implemented pre-trade risk checks and stress-tested 5.4M+ synthetic NSE operations, achieving ∼54K ops/sec with 3.2μs median and 224.7μs p99 latency.
- **Backend Development:** Built backend services using Node.js and Express.js with REST APIs, authentication, and authorization; designed PostgreSQL schemas and SQL queries, and implemented Redis caching, Pub/Sub, WebSockets, and Apache Kafka for real-time and event-driven processing.

DBuddy - AI-Powered Database Management Tool | *Python, PostgreSQL, Gemini API, LangChain* GitHub

- **LLM & Natural Language to SQL:** Integrated Gemini API and LangChain to convert natural-language instructions into SQL/DDL, generate db schemas, and explain query results through an AI-powered db interface.
- **RAG & Intelligent Data Generation:** Developed constraint-aware mock-data generation using LangGraph while preserving foreign-key and unique-key relationships.
- **Database Summarization:** Built an NLP-based summarization feature to generate concise summaries of the user-created databases and their generated content.

RL-Based Crypto Trading Optimization | *Python, FinRL, PPO, LSTM, Gymnasium* GitHub

- **Architecture & State Modeling:** Integrated Recurrent PPO with LSTM memory in FinRL using a **1,103-dimensional state space** with a 50-step lookback across 11 technicle indicators including RSI, MACD.
- **Data Engineering:** Built a pipeline processing **420,000+ 5-minute OHLCV candles** from Binance with transaction fees, bid-ask spreads, quadratic slippage, and maximum drawdown circuit breakers.
- **Reward Engineering:** Designed a multi-objective risk-adjusted reward using rolling-volatility scaling , turnover penalties, and drawdown mitigation, improving test-set returns by **+7.8%** while reducing max drawdown.

Technical Skills

Languages & Tools: C++, SQL, JavaScript, Python, Git, GitHub, Postman
Software Engineering: OOPs, SOLID Principles, System Design, Design Patterns, SDLC, DBMS
AI/ML & Deep Learning : Pandas, NumPy, XGBoost, Neural Networks, RNN, LSTM, GRU, Transformers
Generative AI: LLMs, Prompt Engineering, Gemini API, LangChain, RAG, Embeddings, Vector Databases
Backend & APIs: Node.js, Express.js, REST APIs, WebSockets, JWT Authentication, Redis Pub/Sub, Apache Kafka
Databases & Cloud: PostgreSQL, MongoDB, Redis, Docker, AWS EC2
Coursework: Data Structures & Algorithms, Machine Learning, Reinforcement Learning, Software Engineering

Languages

English (Fluent) | Gujarati (Native) | Hindi (Fluent) | Spanish (Basic)

Achievements

- Solved 555+ Data Structures and Algorithms problems on LeetCode, demonstrating strong algorithmic and problem-solving skills.
- Secured a 98.98 percentile in JEE Main.